BE: Modeling and control of a flexible micro satellite

Participants

- Florian Durix
- Ruben Piberne

Supervisor

- Clément ROOS, ONERA Toulouse
- Jean-Marc BIANNIC, ONERA Toulouse

Date :

25th January 2026

Contents

1 Introduction

DEMETER is a CNES micro-satellite dedicated to the observation of ionospheric disturbances related to seismic and volcanic activity. Its flexible appendices and strict pointing requirements make attitude control a challenging task during scientific operations.

The workshop focuses on the modeling and control of the **DEMETER** satellite's AOCS. It covers the development of a detailed open-loop model including flexible dynamics, actuators, sensors, and estimators, followed by the design of several control strategies such as modal, phase-lead, and robust H_∞ control. The proposed control laws are then validated through simulations and robustness analyses to assess stability and pointing performance.

The numerical values that are used in this workshop are detailed in the table below:

Table 1: DEMETER satellite parameters

Parameter	Description	Value
J_s	Satellite inertia	31.38 kg.m^2
ω	Flexible mode natural frequency	2.6268 rad.s^{-1}
ξ	Flexible mode damping ratio	0.00495
α	Coupling coefficient	0.5736
J_r	Reaction wheel inertia	4×10^{-4} kg.m^2
$T_c^{\max}$	Commanded torque saturation	± 0.05 N.m
$\omega_r^{\max}$	Reaction wheel velocity saturation	± 2800 rpm
τ_m	Measurement delay	0.8 s
σ_m^2	Measurement noise variance	10^{-9}
τ_e	Estimator time constant	0.5 s
T_d	Disturbance torque	0.001 N.m
η_0	Initial condition on flexible state η	10^{-4} rad

2 Open-loop modeling and analysis

In this section, an open-loop model of the DEMETER satellite is developed to analyze its dynamical behavior and highlight the main control challenges. A single-axis linear model is considered, incorporating rigid-body dynamics, flexible appendices, actuator dynamics, and sensor and estimator effects. Frequency- and time-domain analyses are performed to assess stability, controllability, observability, and the impact of disturbances, noise, and structural flexibility.

Satellite Model: The input of the system is the applied torque T, and the output is the satellite pointing angle θ. Equation (1) is a second-order differential equation, therefore the system is of order 2. We define the state variables as $x_1 = \theta$ and $x_2 = \dot{\theta}$, giving the state vector $X = \begin{bmatrix}\theta \\ \dot{\theta}\end{bmatrix}$ and thus $\dot{X} = \begin{bmatrix}\dot{x_1} \\ \dot{x_2}\end{bmatrix} = \begin{bmatrix}x_2 \\ \frac{T}{J_s}\end{bmatrix}$, and $y = x_1$, so the state-space representation is:

$$\dot{X} = \begin{bmatrix}0 & 1 \\ 0 & 0\end{bmatrix} X + \begin{bmatrix}0 \\ \dfrac{1}{J_s}\end{bmatrix} T \qquad \theta = \begin{bmatrix}1 & 0\end{bmatrix} X \tag{1}$$

The poles of the system are the eigenvalues of the state matrix A, given by $\det(\lambda I - A) = 0 \Rightarrow \lambda_1 = 0 \quad \text{and} \quad \lambda_2 = 0$. The system has two integrators (2 poles at 0, hence NOT stricly negatives) and is then unstable (limit of stability).

Taking the Laplace transform of equation (1) yields to:

$$G(s) = \frac{\Theta(s)}{T(s)} = \frac{1}{J_s s^2} \tag{2}$$

For small and compact satellites, flexible dynamics can often be neglected and a rigid-body model provides a sufficiently accurate description. However, the presence of long and slender appendices introduces significant flexibility that must be taken into account. With a very low damping ratio $\xi = 5 \times 10^{-3}$, pronounced flexibility effects and sustained oscillations of the appendices are expected, strongly influencing the satellite dynamics.

$$\begin{cases} \ddot{\theta} - \ddot{\eta} = \dfrac{T}{J_s} \\ \ddot{\eta} = \alpha\, \ddot{\theta} - 2\xi\omega\, \dot{\eta} - \omega^2 \eta \end{cases} \tag{3}$$

Equation 3 consists of two second-order differential equations. The total order of the system is therefore 4. Using the state vector $X = \begin{bmatrix}\theta & \dot{\theta} & \eta & \dot{\eta}\end{bmatrix}^T$, the state-space representation of equation 3 is:

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \dfrac{\omega^2}{1-\alpha} & \dfrac{2\xi\omega}{1-\alpha} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\dfrac{\omega^2}{1-\alpha} & -\dfrac{2\xi\omega}{1-\alpha} \end{bmatrix} X + \begin{bmatrix} 0 \\ \dfrac{1}{J_s(1-\alpha)} \\ 0 \\ \dfrac{\alpha}{J_s(1-\alpha)} \end{bmatrix} T \qquad \theta = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix} X \tag{4}$$

The eigenvalues of the system consist of two poles at the origin, associated with the rigid-body dynamics, and a pair of complex conjugate poles corresponding to the flexible mode. After computation, the flexible poles exhibit a very low damping ratio of $\xi = 7.58 \times 10^{-3}$ at $\omega = 4.02$ rad/s, indicating poorly damped dynamics and pronounced oscillatory behavior. The presence of the two zero poles places the open-loop system at the limit of stability.

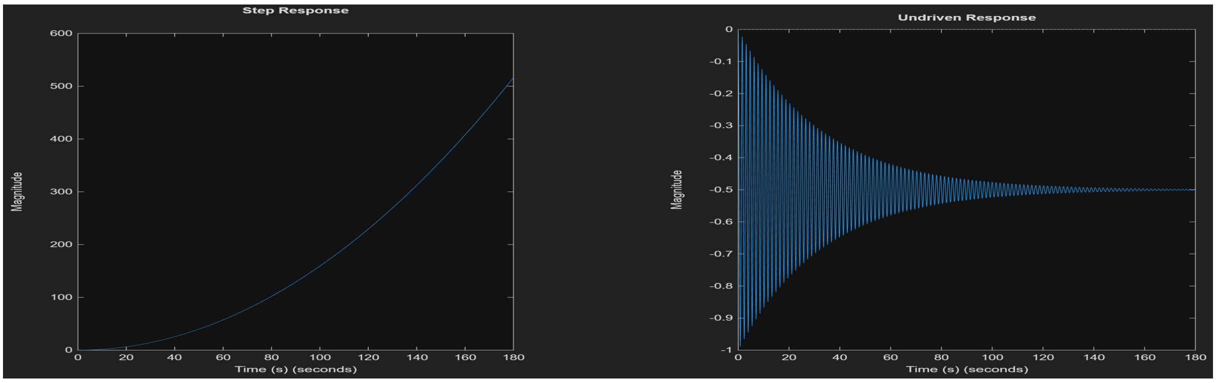

Figure 1: Step and Undriven Response ($\eta_0 = 0.5$)

A step input corresponds to a constant applied torque, resulting in a constant angular acceleration and thus a quadratic growth of the pointing angle, as observed in the step response. This behavior reflects the double-integrator nature of the rigid-body dynamics, leading to an ever-increasing rotation rate. Small oscillations appear during the first seconds, caused by the excitation of the flexible appendices. In the undriven response, the satellite exhibits sustained oscillations due to an initial condition on the flexible mode, with very slow decay. This confirms the very low damping predicted by the pole analysis and highlights the poor natural damping of the flexible dynamics.

The Bode and Nichols diagram of the open-loop system were also plotted to assess its frequency response:

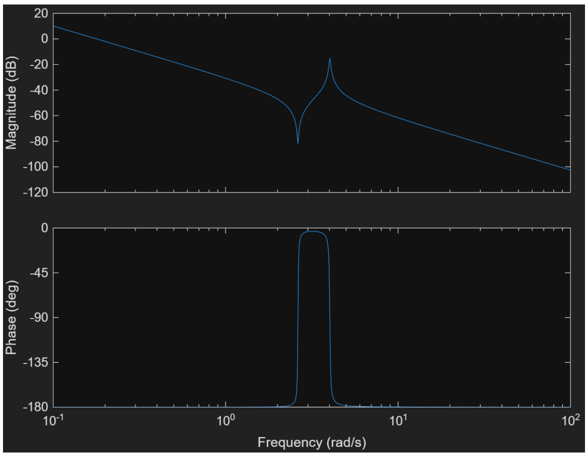

Figure 2: Bode diagram of the open-loop system

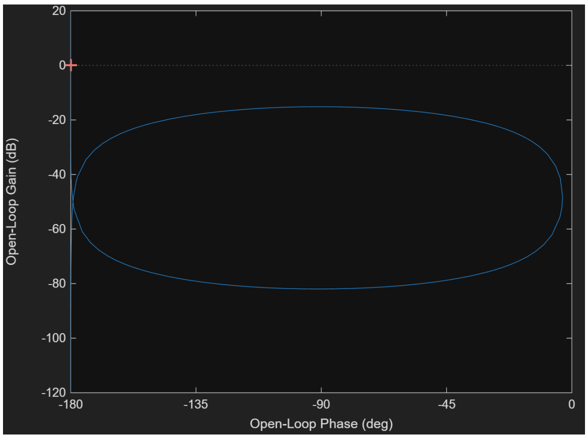

Figure 3: Nichols diagram of the open-loop system

At low frequency, the Bode magnitude exhibits a -40 dB/dec slope, characteristic of the double integrator associated with the rigid-body dynamics. Poorly damped zeros around 2.63 rad/s and poles near 4.02 rad/s generate clear anti-resonance and resonance peaks, revealing the influence of the flexible mode. The phase drops to $-180°$, leading to a null phase margin and confirming marginal stability.

In the Nichols diagram, the loop-shaped curve reflects these resonant phenomena, while the vertical segments at low and high frequencies correspond to the double-integrator behavior, where the phase remains nearly constant while the gain varies significantly.

Finally, we found that the system is both controllable and observable.

Actuator Model: We will now study the satellite actuator, a reaction wheel, that behaves such as a second order linear low pass filter:

$$H_r(s) = \frac{1.214s + 0.763}{s^2 + 2.400s + 0.763} \tag{5}$$

The poles of $H_r(s)$ are -0.38 and -2.02, the linear actuator acts as a first order system of time constant $\tau = \frac{1}{0.38} = 2.63s$ and with a time response of $3\tau \approx 8s$. When the velocity ω_r reach its saturation threshold, the generated torque is null in steady-state because there is no more acceleration. Hence the satellite becomes uncontrolable which is critical so velocity saturation must absolutely be avoided.

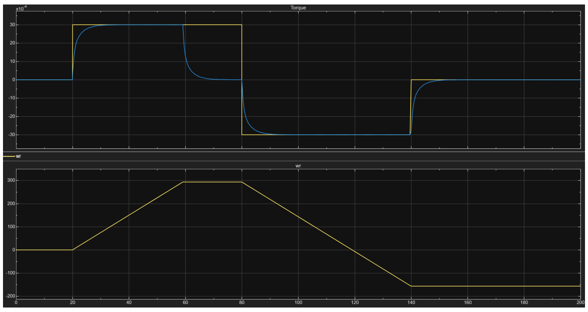

Figure 4: Reaction Wheel Torque and Angular Velocity Responses

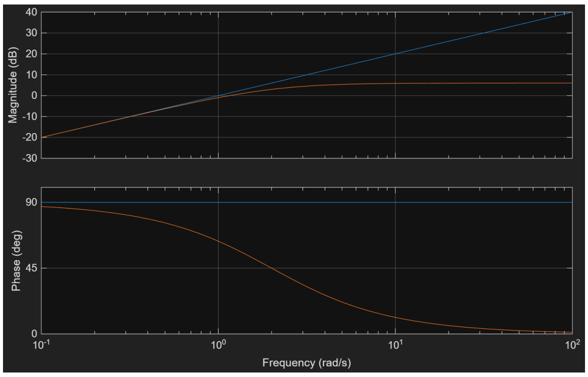

Figure 5: Bode Diagram of the Estimator

Sensor and Estimator Models: The next step is to implement the sensor that measures the satellite's pointing angle θ. It is done by a star tracker that introduces delay τ_m and corrupts the measurement with noise $b(t)$: $\theta_m = \theta(t - \tau_m) + b(t)$. The estimator performs a derivation (differentiation) of the measured pointing angle θ_m in order to reconstruct the angular velocity $\dot{\theta}$, which is not directly measured. Since an ideal derivator would strongly amplify the measurement noise, a **first-order high-pass filter** with time constant τ_e is used instead to add a pole to limit the gain at high frequencies.

Complete Open-Loop Model: The complete open-loop model includes the rigid–flexible satellite dynamics, the reaction wheel model, and the sensor and estimator dynamics. An external disturbance torque T_d, representing environmental perturbations, is applied to the satellite such that $T = T_r + T_d$. A time-domain simulation over 200 s was performed, and the results are shown in Fig. 7. The commanded torque T_c is compared with the actual torque T, the measured angle θ_m with the true angle θ, and the estimated angular velocity $\dot{\theta}_e$ with $\dot{\theta}$.

The behavior of the applied torque T is governed by the actuator dynamics described in the **Actuator Model** section. For a positive torque command, the reaction wheel angular velocity ω_r increases linearly until it reaches the saturation limit $\omega_r^{\max}$, at which point the torque drops to zero. When T_c becomes negative, T follows with a finite time response of 8s and ω_r decreases, becoming negative, but without reaching $-\omega_r^{\max}$, therefore, no additional saturation occurs. At $t = 100\,\mathrm{s}$, the disturbance torque T_d is introduced, resulting in a steady-state error on the applied torque.

The measured pointing angle θ_m closely matches the true angle θ, with the measurement delay being only slightly visible and the noise remaining negligible. In contrast, a larger discrepancy is observed between the estimated angular velocity $\dot{\theta}_e$ and the true velocity $\dot{\theta}$, due to the estimator dynamics, the star tracker delay, and measurement noise, which is amplified by the derivation (at $t = 160\,\mathrm{s}$). The angular velocity signals are also slightly affected by oscillations induced by the flexible mode.

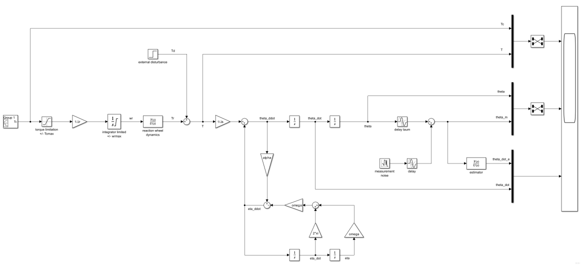

Figure 6: Open-Loop System Simulink

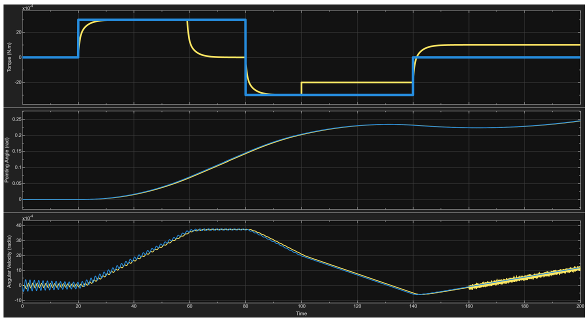

Figure 7: Time Domain Simulation of the Open-Loop on 200s

3 Closed-loop control

In this part, small pointing angles are considered. Closed-loop control is achieved using PD or PID controllers, designed through different synthesis approaches: **Modal**, **Phase-Lead** and $\mathbf{H}_\infty$.

3.1 Modal Control

The objective of modal control is to impose a desired closed-loop dynamic behavior by specifying target closed-loop poles. A PD or PID controller are then tuned to approximate this behavior, within the limitations of its structure.

Open loop Computation The system time response is mainly governed by its dominant poles, i.e. the slowest poles, which are the closest to the imaginary axis. By removing all the nonlinearities of the satelitte model, we obtain the following open loop pole described in Tab.2.

The number of poles that can be fixed are directly given by the following rules: *The number of chosen poles is equal to the number of independent measures q.*

Poles	Damping	Freq (rad/s)	Meaning
0.00	-1.00	0.00	Satellite Core Integrator
0.00	-1.00	0.00	Satellite Core Integrator
-0.377	1.00	0.377	Reaction Wheel
-2.02	1.00	2.02	Reaction Wheel
-2.00	1.00	2.00	Estimator
$-0.0305 + 4.02i$	7.58×10^{-3}	4.02	Flexible Mode
$-0.0305 - 4.02i$	7.58×10^{-3}	4.02	Flexible Mode

Table 2: Pole of the system and dynamical characteristic associated

In our case, we have two **independent** measurements, respectively θ and $\dot{\theta}$ (cf. Fig.8). So we can fix 2 poles regarding the **eigenstructure assignment algorithm**.

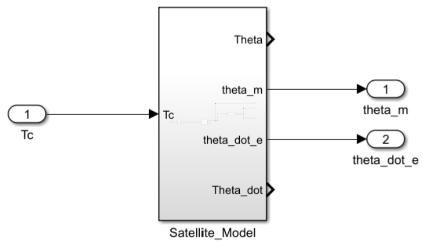

Figure 8: Open loop Simulink Representation

Design of a PD controller The objective of the modal control strategy is to impose a desired closed-loop dynamic behavior. In this study, the satellite closed-loop response is required to be dominated by a well-damped second-order behavior with a damping ratio of $\delta = 0.7$. The response should also be as fast as possible, with a natural frequency ω_n chosen as large as possible within actuator limitations.

As explained previously, a PD controller is designed to approximate the desired modal behavior. The **target closed-loop dynamics is defined by a second-order characteristic polynomial**:

$$p(s) = s^2 + 2\,\delta\,\omega_n\,s + \omega_n^2.$$

The roots of this polynomial correspond to the desired closed-loop poles. They are computed in matlab using the `roots` function:

```
lambda = roots([1 2δω_n ω_n^2])
```

We now want these poles to be the poles of the closed-loop system. This means that the closed-loop dynamics must admit the imposed values λ_i as eigenvalues.

From eigenvalue theory, an eigenvalue λ_i is associated with a non-zero eigenvector v_i satisfying $A_{cl} v_i = \lambda_i v_i$.

Since the closed-loop poles are the eigenvalues of the closed-loop state matrix, the controller K must be designed such that this condition is satisfied.

In the case of output feedback, this condition can be rewritten as: $(A - \lambda_i I)v_i + Bw_i = 0$.

Where w_i represents the contribution of the control input associated with the eigenvector v_i.

The vectors (v_i, w_i) are computed to ensure that the imposed closed-loop poles are compatible with both the system dynamics and the control input. For each desired pole λ_i, a non-zero solution (v_i, w_i) of the equation $(A - \lambda_i I)v_i + Bw_i = 0$ is obtained by computing the null space of the matrix $[A - \lambda_i I \quad B]$:

```
VW1 = null([sys.A - lambda(1)*eye(n), sys.B]);
VW2 = null([sys.A - lambda(2)*eye(n), sys.B]);
V = [VW1(1:n), VW2(1:n)];
W = [VW1(n+1), VW2(n+1)];
```

The objective is to select the largest possible natural frequency ω_n in order to obtain the fastest closed-loop response, while ensuring that the imposed second-order poles remain dominant. This requires all remaining closed-loop poles to be faster than the dominant poles, i.e. to have a larger magnitude.

In this study, the value of ω_n is selected manually through successive iterations, although this process could be automated. Let $\lambda_{d,i}$ denote the imposed dominant closed-loop poles and λ_k the remaining closed-loop poles. For each candidate value of ω_n, the closed-loop poles are computed and the dominance condition:

$$\forall\, \lambda_k \notin \{\lambda_{d,i}\}, \quad |\lambda_{d,i}| < |\lambda_k|$$

is verified by comparing the pole magnitudes. This approach allows the determination of the maximum admissible value of ω_n while preserving the dominance of the desired closed-loop dynamics.

The initial value of ω_n is selected below the reaction wheel actuator bandwidth (0.370 rad/s), which is known to be a physical limitation of the system. Figure 9 illustrates why this bandwidth represents the limiting factor, and ω_n is then reduced step by step to satisfy the closed-loop requirements.

Finally, a value of $\omega_n = 0.180$ rad/s is selected. This value provides the fastest achievable response while preserving the dominance of the imposed second-order dynamics.

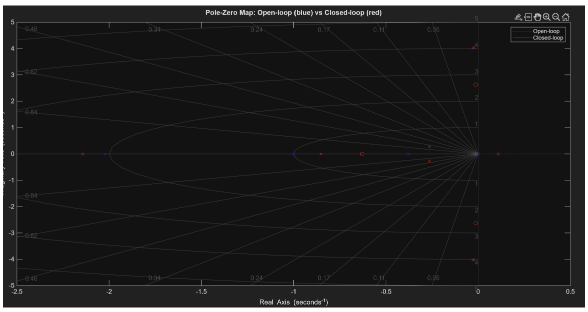

Figure 9: Unstable Closed Loop with $\omega_n = 0.370$

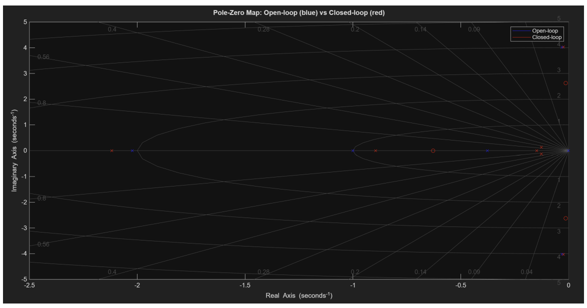

Figure 10: Stable closed loop with $\omega_n = 0.180$

Phase & Gain margin From the Bode diagram, the gain and phase margins are extracted. The controller show satisfactory robustness, with a gain margin $\Delta_G \in [6\,\text{dB},\ 12\,\text{dB}]$ and a phase margin $\Delta_\phi \approx 45°$.

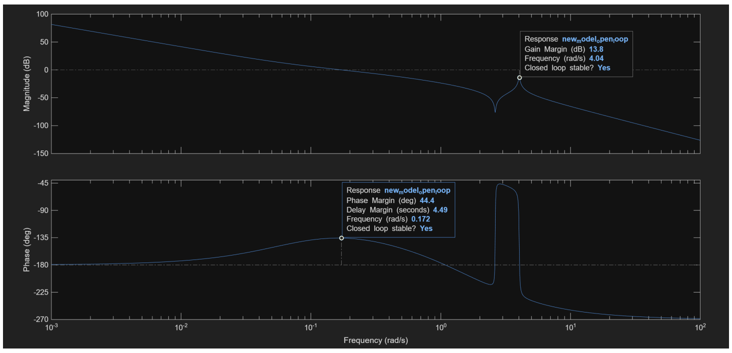

Figure 11: Gain and Phase margin computed from matlab using the Bode Diagram

.

Computation of the Feedforward gain H: In order to achieve zero steady-state error, the closed-loop output must satisfy $\theta = \theta_{\text{ref}}$. Given the closed-loop architecture $\theta(s) = H\,G_{\text{cl}}(s)\,\theta_{\text{ref}}(s)$, this condition requires that the static gain of the overall closed-loop system be equal to unity: $\lim_{s\to 0} H\,G_{\text{cl}}(s) = 1$. Consequently, the feedforward gain H is chosen as the inverse of the DC gain of the closed-loop transfer function, $H = \frac{1}{\lim_{s\to 0} G_{\text{cl}}(s)} = \frac{1}{\texttt{dcgain}(G_{\text{cl}})}$. We compute H=0.3739

Simulation A 200 s closed-loop simulation is first performed to evaluate the nominal behavior of the controller. The disturbance torque and star tracker noise are applied at $t = 100\,\text{s}$ and $t = 160\,\text{s}$, respectively. The corresponding pointing angle response is shown in Fig. 13.

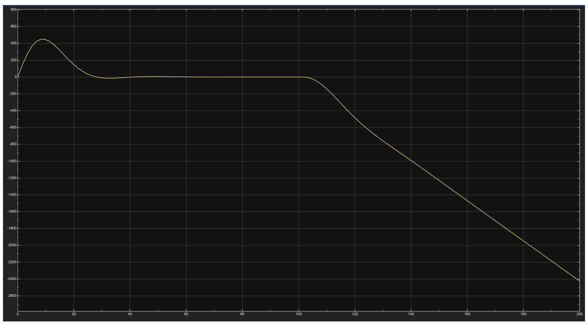

Figure 12: 200s simulation - Reaction wheel angular velocity for $\theta_{\text{ref}} = 0.5°$

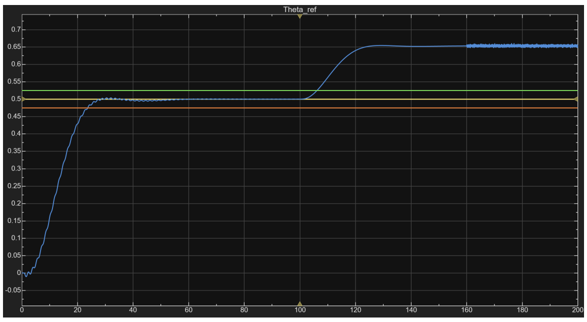

Figure 13: 200s simulation - Nominal closed-loop response for $\theta_{\text{ref}} = 0.5°$

For a reference pointing angle of $\theta_{\text{ref}} = 0.5°$, the system reaches the desired value after approximately 20 s. The response exhibits a very small overshoot, remaining below 5% of the reference angle.

At $t = 100\,\text{s}$, the application of a disturbance torque causes the pointing angle to increase up to approximately 0.625°. This perturbation persists until $t = 160\,\text{s}$, when star tracker noise is introduced on the measured pointing angle. Despite the presence of measurement noise, the pointing angle remains bounded within a narrow range, indicating good robustness to noise. However, the sustained deviation induced by the disturbance torque reveals limited disturbance rejection capability.

Pointing Angle Comparison and Physical Limitations To further assess the controller performance, different operating conditions are considered. A larger reference angle $\theta_{\text{ref}} = 10°$ is then applied. In this case, the reaction wheel reaches its saturation limit at 2800 rpm, as illustrated in Fig. 14. Although the pointing angle response remains acceptable in simulation, the actuator saturation indicates that the controller exceeds physical limitations and is therefore not implementable in practice.

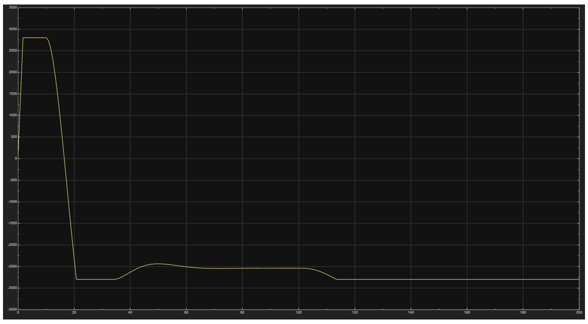

Figure 14: 200s simulation - Reaction wheel angular velocity for $\theta_{\text{ref}} = 10°$

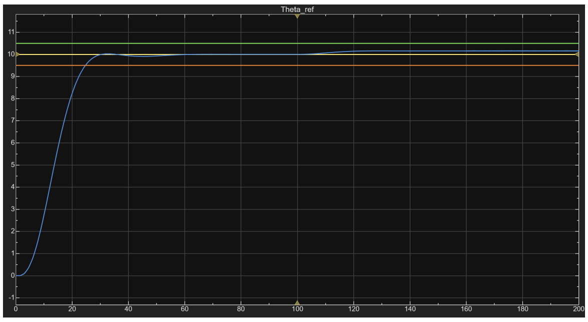

Figure 15: 200s simulation - Nominal closed-loop response for $\theta_{\text{ref}} = 10°$

Finally, as required, a maximum admissible pointing angle of approximately $\theta \approx 3.15°$ is computed. This value represents the largest reference angle that can be tracked without immediately reaching actuator saturation.

The reference is then reset to $\theta_{\text{ref}} = 0.5°$ and the simulation duration is extended to 400 s. Despite the reduced reference angle, the reaction wheel again reaches saturation over long simulation times, indicating that the controller is not suitable for long-duration operations.

Integrator augmentation In order to improve steady-state accuracy and reject constant input disturbances, an integrator is added to the control loop. This guarantees zero steady-state error on the regulated output.

The integrator state is defined as:

$$\dot{x}_i = -y,$$

which leads to an augmented open-loop system of order $n + 1$.

The augmented state vector and state-space representation is then given by:

$$X = \begin{bmatrix} x \\ x_i \end{bmatrix}, \quad \dot{X} = \underbrace{\begin{bmatrix} A & 0 \\ -P & 0 \end{bmatrix}}_{A_{\text{aug}}} X + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{B_{\text{aug}}} u, \quad Y = \underbrace{\begin{bmatrix} 0 & -1 \\ -C & 0 \end{bmatrix}}_{C_{\text{aug}}} X.$$

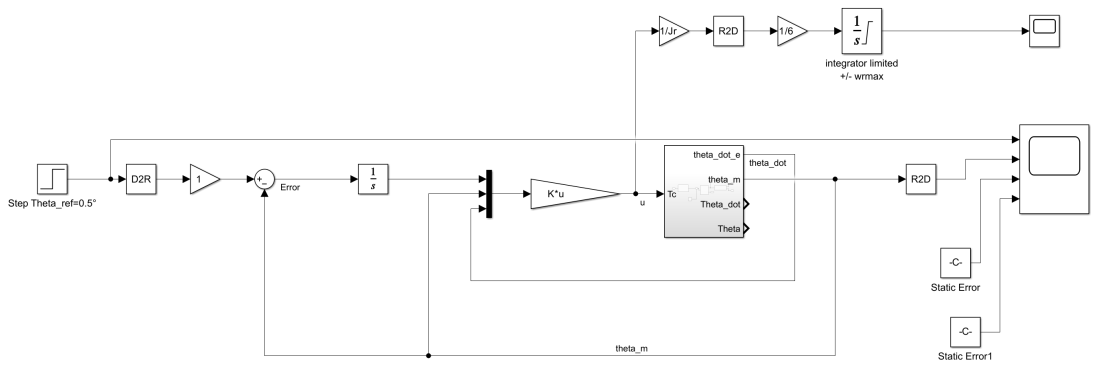

Figure 16: PID Controller implemented on Matlab/Simulink

Since the system order is increased by one due to the integrator, an additional closed-loop pole can be assigned. Following the same modal control strategy as previously, this extra pole is chosen as:

$$\lambda_3 = -\omega_n$$

That will ensure a stable integrator dynamics while preserving the dominance of the desired second-order behavior.

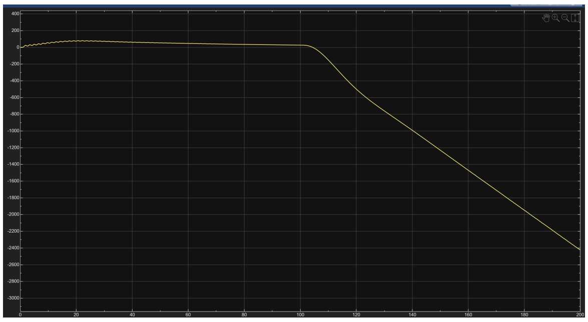

Figure 17: Time evolution of the reaction wheel velocity for 200s simulation

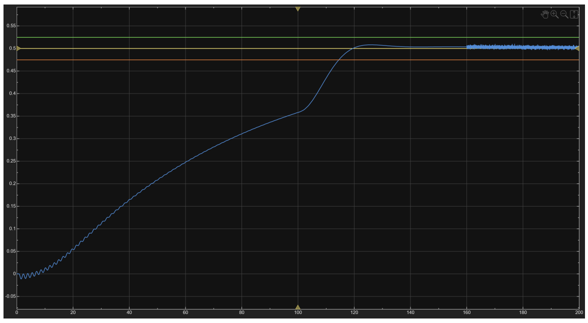

Figure 18: Time Response of the PID controller for 200s simulation

3.2 Phase-Lead Control

A phase-lead control is applied in this section. The closed-loop structure is presented in the Fig.19. The feedback is only made using θ_m. The objective will be to tune the phase-lead controller $K_1(s) = k\frac{1+a\tau_1 s}{1+\tau_1 s}$, the PI controller $K_2(s) = \frac{1+\tau_2 s}{\tau_2 s}$ and the feedforward gain H.

3.2.1 Design of the Phase-Lead Controller $K_1(s)$

The first step of the phase-lead control design is to focus on the phase-lead controller $K_1(s)$ only, hence, $K_2(s) = 1$ for this part.

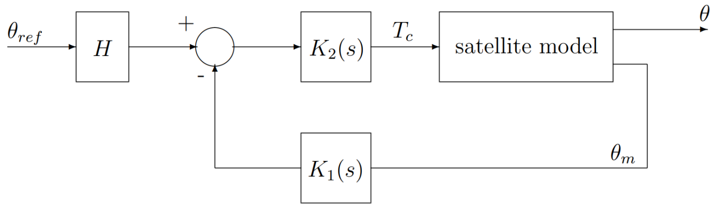

Figure 19: Closed-Loop Interconnection for Phase-Lead Control

Influence of k: We set $\tau_1 = 0$ and $a = 0$, which yields a pure proportional controller $K_1(s) = k$. Whatever the value of k, the closed-loop interconnection remains unstable. The root locus shows that the two poles at the origin, corresponding to the double integrator dynamics, move along the imaginary axis and never enter the left-half plane as k varies. From the Bode phase diagram, the phase is already equal to or below $-180°$ at low frequencies. Since a constant gain does not modify the phase, the phase margin remains negative for all values of k. Therefore, additional phase must be introduced using a phase-lead controller in order to increase the phase margin and shift the closed-loop poles into the stable region.

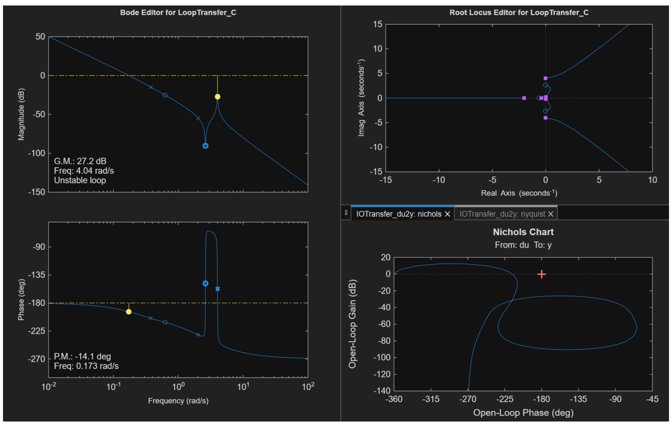

Figure 20: Open-Loop Root Locus, Bode and Nichols Diagram using *sisotool* function

Tuning of a, k and τ_1 for $K_1(s)$: The next step consists in tuning the parameters of the phase-lead controller $K_1(s)$ in order to achieve a target phase margin $\phi_{\text{target}} = 50°$. From the open-loop analysis (Fig. 20), the gain crossover frequency is $\omega_c = 0.173\,\text{rad/s}$ and the phase margin is $\phi_m = -14.1°$. Therefore, an additional phase lead of $\phi = \phi_{\text{target}} - \phi_m = 64.1°$ is required.

The parameters a and τ_1 of the phase-lead controller are computed using the standard relations:

$$a = \frac{1+\sin\phi}{1-\sin\phi} = 18.91 \qquad \tau_1 = \frac{1}{\omega_c\sqrt{a}} = 1.39\,\text{s}. \tag{6}$$

The introduction of the phase-lead controller modifies the loop gain and shifts the crossover frequency. In order to preserve the desired crossover frequency ω_c, the gain k is adjusted such that the magnitude of the open-loop transfer function at ω_c remains equal to 0 dB: $|OL(j\omega_c)| = |G(j\omega_c)K_1(j\omega_c)| = 1$. Since the gain of the plant $G(j\omega_c)$ is already fixed, this condition reduces to $|K_1(j\omega_c)| = 1$, which yields:

$$k = \sqrt{\frac{1 + (\tau_1\omega_c)^2}{1 + (a\tau_1\omega_c)^2}} \quad \Rightarrow \quad k = \frac{1}{\sqrt{a}} = 0.2304. \tag{7}$$

An other way to compute k is to implement the controller in the *sisotool* function and look at the gain at $\omega = 0.173\text{rad/s}$, $G_{0.173\text{rad/s}} = -12.7\text{dB}$. Then, $k = 10^{\frac{-12.7}{20}} = 0.231$.

Feedforward gain H: In order to achieve zero steady-state error we compute $H = \frac{1}{4.34} = 0.2305$ based on the explanation in **Modal Control** section.

Time-Domain Response: The feedforward gain H and the phase-lead controller $K_1(s)$ were implemented in the closed-loop system. A time-domain simulation of 200 s was performed to evaluate the pointing angle response θ to a reference step of 0.5°. The corresponding results are shown in Fig.21 and Fig.22. The closed-loop system is stable and achieves the targeted phase margin of $\phi = 50°$. The steady-state tracking error is quasi-null in nominal conditions. However, the closed-loop dynamics remain very slow, with a response time of approximately 58.8 s. In addition, the disturbance rejection performance is poor: the applied torque disturbance induces a steady-state pointing error of $\varepsilon = 0.752°$, corresponding to about 150% of the reference step amplitude. No measurement noise is present at $t = 160$ s in this simulation.

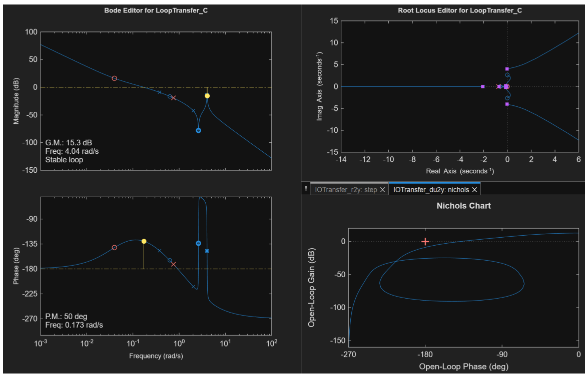

Figure 21: Open-Loop Root Locus, Bode and Nichols Diagram using *sisotool* function after $K_1(s)$ implementation

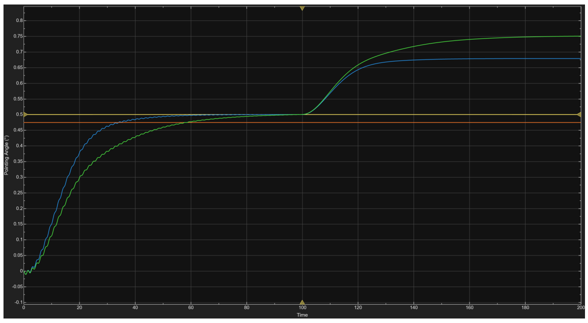

Figure 22: Time Domain Simulation of the Closed-Loop with $K_1(s)$ (green) and $K_{1,new}(s)$ (blue) on 200s

$K_1(s)$ refinement: Because of the poor performance related to the first tuning of the phase-lead controller regarding time response, we want to improve it by moving the pole and the zero of $K_1(s)$ to find new parameters that still ensure a phase-margin of 50°.

$$K_1(s) = k_{\text{new}}\frac{s - z_{\text{new}}}{s - p_{\text{new}}} = k_{\text{new}}\frac{1 + a_{\text{new}}\tau_{1,\text{new}}s}{1 + \tau_{1,\text{new}}s} \Rightarrow \tau_{1,\text{new}} = -\frac{1}{p_{\text{new}}}, \quad a_{\text{new}} = -\frac{1}{\tau_{1,\text{new}}z_{\text{new}}} \tag{8}$$

By using the *sisotool* function, the phase-lead controller zero and pole were adjusted to $z_{\text{new}} = -0.0579$ and $p_{\text{new}} = -1.47$, respectively. This yields the updated parameters $\tau_{1,\text{new}} = 0.6803\,\text{s}$ and $a_{\text{new}} = 25.39$. Applying the same gain-tuning procedure as previously, the new gain was obtained as $k_{\text{new}} = 0.3203$, which also defines the updated feedforward gain $H_{\text{new}} = 0.3203$.

The corresponding time-domain response is shown in Fig. 22. Compared to the previous design, the closed-loop system is faster (with a response time of approximately 34.1 s) and exhibits a reduced steady-state error in the presence of the disturbance torque.

Phase-Lead Controller in Direct Link: We assessed the impact of implementing $K_1(s)$ in the direct link (Fig. 23) instead of the feedback link. We can observe an overshoot of the pointing angle (Fig. 24).

From a theoretical viewpoint, the phase-lead controller K_1 is characterized by a low-frequency zero. When $K_1(s)$ is placed in the feedback path, this zero does not appear in the numerator of the closed-loop transfer function from θ_{ref} to θ. However, if $K_1(s)$ is implemented in the direct (forward) path, the zero is included, which introduces an overshoot in the time response.

From a more intuitive perspective, $K_1(s)$ acts like a proportional-derivative controller: the derivative-like action estimates $\dot{\theta}$ to improve stability. When applied in the forward path to a step input, this pseudo-derivative amplifies the initial response, producing a large overshoot.

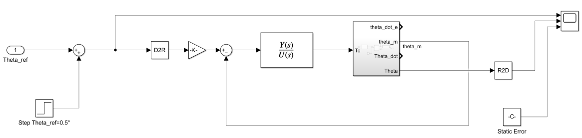

Figure 23: $K_1(s)$ in Direct Link Simulink

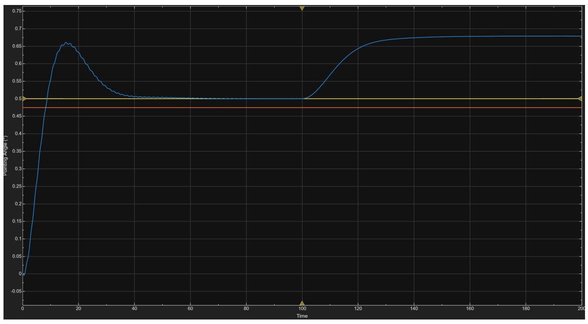

Figure 24: Time Domain Simulation of the Closed-Loop with $K_1(s)$ in Direct Link

3.2.2 Design of the PI Controller $K_2(s)$

In this section, we want to tune the PI controller $K_2(s)$ in order to have a better rejection of torque disturbance.

PI controller or pure integrator: A pure integrator $1/s$ ensures zero steady-state error, but it introduces a $-90°$ phase lag, which reduces the phase margin and can destabilize the system, especially in combination with the phase-lead controller in feedback. By adding a zero with the PI ($K_2(s) = K_p + K_i/s$), the low-frequency zero partially compensates the integrator's phase lag, improving the phase margin and preserving stability while still providing disturbance rejection.

Tuning of τ_2: The PI controller has the following transfer function, $K_2(s) = \frac{1+\tau_2 s}{\tau_2 s}$. We want to find the value of τ_2 to have a phase margin equal or lower to 45°. The open-loop $L(s) = K_2(s)G(s)K_1(s)$ is considered, to meet the requirements, we want $\arg[L(j\omega_c)] \geq -180° + 45° = -135°$. The requirement on τ_2 is:

$$\arg[L(j\omega_c)] = \arg[K_1(j\omega_c)G(j\omega_c)] + \arg[K_2(j\omega_c)] = \Phi_1 + \arctan(\omega_c \tau_2) - 90° \tag{9}$$

$$\tau_2 \geq \frac{1}{\omega_c} \tan(\Phi_1 - 45°) \tag{10}$$

The phase-margin that we previously set with the phase-lead controller was $\phi = 50.6 \Rightarrow \Phi_1 = -129.35°$. At $\omega_c = 0.173$rad/s, we can compute $\tau_2 = 58.46s$ which is the optimal value to get the fastest PI controller while ensuring a phase-margin superior to 45°.

Time domain simulation with PI and Phase-Lead controllers: We performed a new simulation implementing both the PI and phase-lead controllers to assess their combined performance. The results are shown in Fig. 25.

We observe that the system response is faster than before, with a time response of 27.81 s prior to the torque disturbance. The torque perturbations are effectively rejected; however, the pointing angle does not return within the 5% band until $t = 228.4$ s, i.e., 128.4 s after the disturbance.

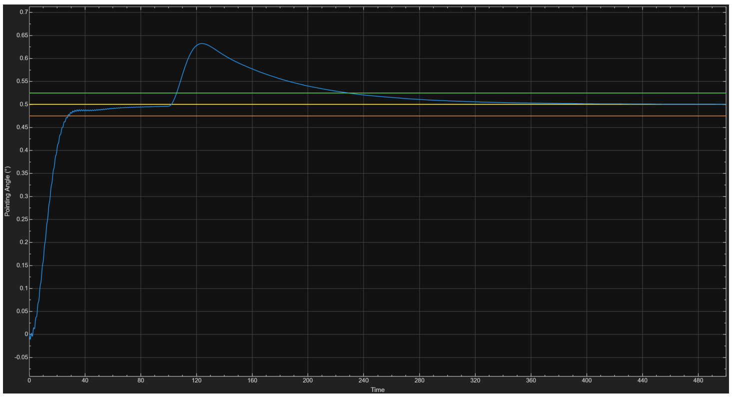

Figure 25: Time-Domain Simulations of the Closed-Loop System with PI and Phase-Lead Controllers during 500s

3.3 H_∞ Control

H_∞ control is applied to the system. Full-order controllers are initially designed using the `hinfsyn` function from the MATLAB Robust Control Toolbox. The design aims to achieve low steady-state error, maintain reasonable control signal amplitudes, ensure effective disturbance rejection, provide accurate tracking performance, and attenuate measurement noise.

Tuning of the sensitivity function S(s) In this part, a weighting function $W_1(s)$ is defined to shape the sensitivity function $S(s)$ and reduce the steady-state tracking error. From an $\mathcal{H}_\infty$ perspective, the objective is to enforce $\|W_1(s)\, S(s)\|_\infty < 1$.

The following constraints are imposed on the sensitivity function:

- $|S(0)| < 10^{-3}$ that ensures a very small steady-state error. A strictly zero value is not imposed to avoid a non-realizable weighting function.
- $\forall\, \omega < \omega_c = 0.18$ rad/s, $|S(j\omega)| < 1$ that guarantees good disturbance rejection at low frequencies, compatible with the actuator dynamics. The cutoff frequency ω_c is chosen close to one half of the actuator bandwidth to avoid excitation of unmodeled or saturating dynamics.
- $\forall\, \omega \geq \omega_c, |S(j\omega)| < 2$ that limits sensitivity peaking, ensuring adequate gain and phase margins and preserving robustness. It corresponds to a disk margin greater than 0.5, leading to a sensitivity function smaller than $\frac{1}{0.5} = 2$.

In matlab we defined the given gabarit and we computed the associated weighting function W1(s):

```
lambda = makeweight([0.001, [wc 1], 2]);
W1 = minreal (1/WS);
```

Compute a Controller K that satisfy the requirements By fixing $W_2(s) = 1$, we first computed a controller K with 8 states, which is of higher order than the plant.

After the computation, a value of $\gamma = 1.45$ was obtained. However, this value does not satisfy the required closed-loop stability margins. In fact, From a frequency-domain perspective, γ is directly related to the robustness margins of the closed-loop system. In particular, the minimal distance D_m between the Nyquist plot of the open-loop transfer function and the critical point $(-1, 0)$ satisfies $D_m = \frac{1}{\gamma}$.

This means that the weighting function $W_2(s) = 1$ imposed too many constraints on the system. An iterative tuning procedure was therefore applied to determine the largest admissible value of $W_2(s)$ such that the condition $\gamma \leq 1$ is satisfied. This process led to $W_2(s) = 0.2190$, for which $\gamma = 0.97528$, which complies with the requirements.

Figure 26 compares the magnitudes of the sensitivity function $S(s)$ and of $(G(s)K_{H_1}(s))^{-1}$ with the inverse weighting function $W_1^{-1}(s)$. It can be observed that $|S(j\omega)|$ remains below $|W_1^{-1}(j\omega)|$ over the entire low-frequency range, which confirms that the static error and low-frequency disturbance rejection requirements are satisfied. Although $|(GK_{H_1})^{-1}|$ exceeds the bound at high frequencies, this does not violate the design objective, since $W_1(s)$ is intended to constrain the low-frequency behavior of the sensitivity function.

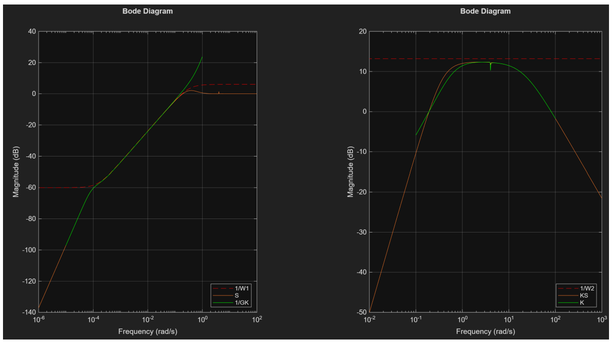

Figure 26: Comparison of Magnitudes

Fig.26 compares the magnitudes of $K_{H_1}(s)S(s)$ and $K_{H_1}(s)$ with $W_2^{-1}(s)$. The controller magnitude remains below the imposed constraint, ensuring a proper high-frequency roll-off and limiting the amplification of measurement noise. Instead of applying $W_2(s)$ directly to the actuator torque, it is placed upstream of the reaction wheel model. This choice exploits the inherent low-pass behavior of the actuator, which naturally attenuates high-frequency components due to its excess of poles over zeros. As a result, a constant weighting function $W_2(s)$ is sufficient to meet the robustness and noise attenuation requirements.

Table 3: Poles of $KH_1(s)$

Pole	Damping ζ	Frequency (rad/s)
-1.56×10^{-4}	1.00	1.56×10^{-4}
$-0.509 \pm 0.231\,i$	0.911	0.559
-1.94	1.00	1.94
-2.00	1.00	2.00
$-0.0377 \pm 4.02\,i$	9.37×10^{-3}	4.02
-31.5	1.00	31.5

Table 4: Zeros of $KH_1(s)$

Zero
$-0.0305 \pm 4.0226\,i$
-2.0228
-2.0000
-0.3772
0
0

The poles of $KH_1(s)$ all have negative real parts, which confirms the internal stability of the controller. One pole is located very close to the origin ($\approx 10^{-4}$), meaning that the controller exhibits an integrator-like behavior and thus provides low-frequency gain.

However, the controller also contains zeros very close to the origin, which partially cancel this integrating effect. As a result, the effective low-frequency gain of the loop is reduced, leading to limited disturbance rejection performance.

This behavior is confirmed by the Bode diagram (cf Fig.27): at low frequencies (where external disturbances are dominant), the loop gain is relatively small, resulting in a non-negligible steady-state error. Consequently, despite stability, the disturbance rejection capability of the system remains poor.

Comparison between the H_∞ Controller and the PD Controller As we can observe on the Fig.27, both controllers have very similar frequency responses. Since the high-frequency roll-off of the H_∞ controller is not relevant here, it can be well approximated by a PD controller : $K_{H_1}(s) \approx K_{PD}(s) = K_P + K_D \frac{s}{1+\tau_D s}$ with $K_P = 10^{-\frac{K_{H_1}(0)}{20}} = 0.0002512$, $K_D = 5$ and $\tau_D = \frac{1}{\omega_{\text{roll-off}}} = 1.11\text{s}$.

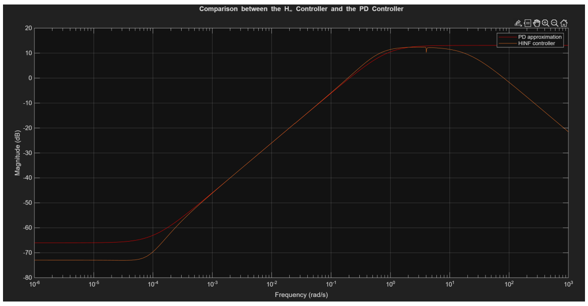

Figure 27: Comparison between the H_∞ controller and its PD approximation

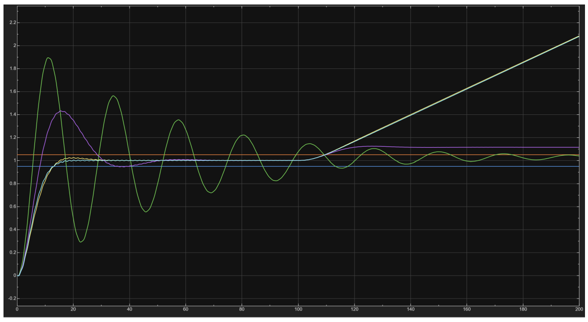

Figure 28: Influence of the proportional gain K_p on the closed-loop time response

Influence of the Proportional Gain K_p on the Closed-Loop Response

- K_{H_∞}: slow response with no overshoot and high steady-state accuracy. However, the disturbance rejection is poor, as the static error increases linearly over time when a disturbance torque is applied.
- K_{H_∞} approximated by a PD controller with $K_p = 0.0002512$: same behavior as the K_{H_∞} controller, which is expected since this controller is a direct approximation of it.
- PD controller with $K_p = 0.5$: good compromise between speed and robustness, with a faster response and improved disturbance rejection, but with some overshoot. The static error remains present but is reduced and stabilizes.
- PD controller tuned with $K_p = 2$: very fast response but strong overshoot and oscillations, indicating an aggressive tuning that leads to a poorly damped system, close to instability.

Torque disturbance rejection: The previous results show that the rejection of torque disturbances must be improved. To reduce the impact of the disturbance torque on the tracking error, a weighting function $W_3(s)$ is introduced in the $\mathcal{H}_\infty$ loop.

The disturbance is deliberately injected *before the actuator* in the control architecture. The reaction wheel dynamics naturally behave as a low-pass filter due to their dominant pole structure. As a consequence, including the actuator dynamics inside $W_3(s)$ is unnecessary and would only increase the order of the generalized plant.

By placing $W_3(s)$ upstream of the actuator, the controller is forced to generate high gain at low frequencies in order to reject constant or slowly varying disturbances, leading to a controller behavior close to that of an integrator while avoiding the introduction of additional states.

The Bode Diagram of the 4-block diagram with K_{H_1} is shown in Fig.29. We observe that the transfer function $T_{w_1 \to z_1}$ acts as a high pass filter, $T_{w_1 \to z_2}$ behaves like the K_{H_1} controller that was designed and $T_{w_2 \to z_1}$ has a very large gain at low frequencies which is responsible for the poorly disturbance rejection.

We set $W_3(s) = 1$ and we obtain the following time-domain results:

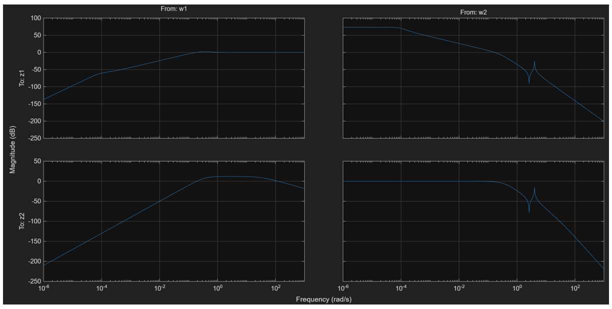

Figure 29: 4-block diagram with K_{H_1} Bode Diagram

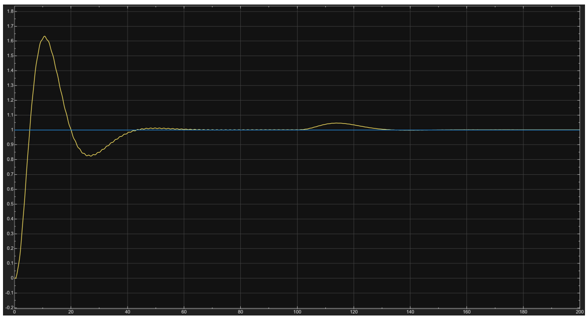

Figure 30: Time-response of Pointing Angle θ with K_{H_2}

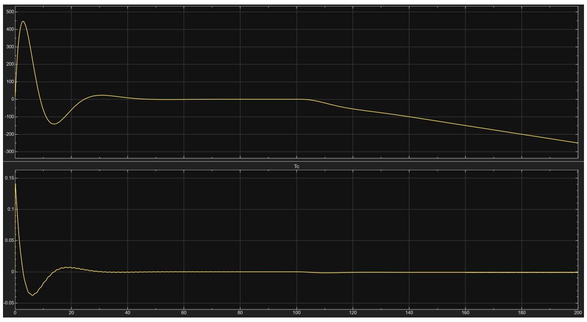

Figure 31: Time-response of Reaction Wheel Commands with K_{H_2}

We observe that the implementation of K_{H_2} is robust to torque disturbances, as the disturbance introduced at $t = 100\,\text{s}$ is rejected within a few seconds. However, the closed-loop response exhibits a significant overshoot, which indicates insufficient damping and degrades the dynamic performance by increasing the settling time and potentially inducing actuator saturation. Although the steady-state accuracy is preserved, the overshoot results in a slower and less well-behaved transient response.

Feedforward Controller A feedforward term was implemented to evaluate its impact on the H_∞ performance of the controller. After computing the resulting controller, a new value of $\gamma = 0.9813$ was obtained. The H_∞ norm was thus reduced from an initial value close to 1.9 to below 1, indicating that the robustness requirements are now satisfied and that adequate stability margins are recovered. This improvement can be explained by the increase in the degrees of freedom of the control structure by directly using the measured reference θ_{ref}, the feedforward action anticipates the system response and reduces the effort required from the feedback loop, thereby improving closed-loop stability and performance.

Noise Attenuation & Variation of r_α and r_ω First, the impact of measurement noise on the closed-loop system was analyzed. As shown in Fig.32, the overall time response remains acceptable in terms of tracking performance. However, the effect of noise is clearly visible on the reaction wheel velocity and on the commanded torque, as illustrated in Fig.33. This indicates that the controller exhibits limited robustness with respect to noise.

In addition, a small variation of the system was introduced by increasing the flexible mode contribution and its associated dynamics. Under this modification, the closed-loop behavior significantly degrades and the system becomes unstable, as shown in Fig.34 and

Fig.35. As a result, the system is no longer operational, highlighting the controller's sensitivity to model variations.

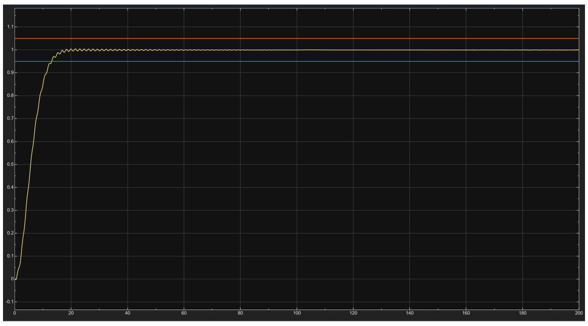

Figure 32: Time Response of the KH3 controller with $r_\alpha = 1; r_\omega = 1$ for 200s simulation

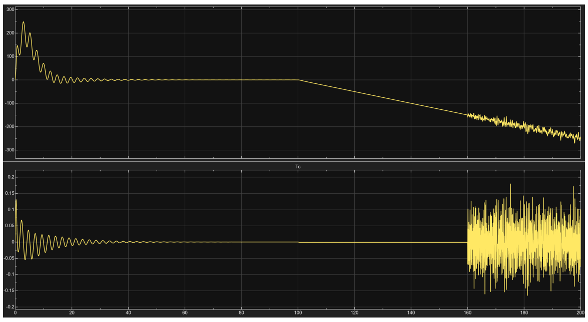

Figure 33: Time evolution of the reaction wheel velocity and the commanded torque for 200s simulation

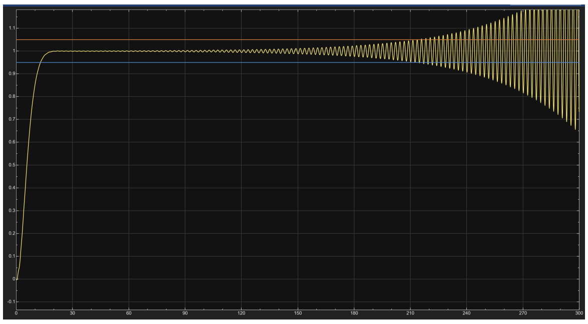

Figure 34: Time Response of the KH3 controller with $r_\alpha = 1.05; r_\omega = 1.05$ for 300s simulation

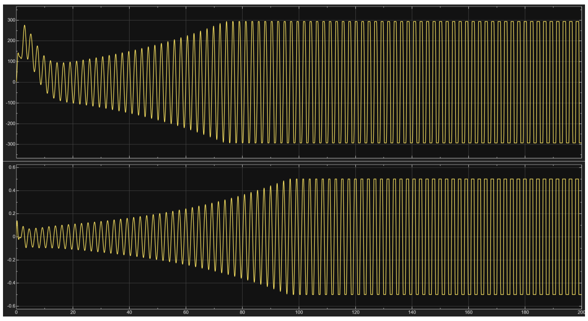

Figure 35: Time evolution of the reaction wheel velocity and the commanded torque for 300s simulation

Noise Rejection: As shown previously, we have a poorly noise rejection on the commanded torque. The magnitude of $T_{w_3 \to z_2}$ (Fig. 36) exhibits a very large gain at high frequencies, reaching values of the order of 80 dB. This behavior indicates a strong amplification of high-frequency noise rather than effective rejection. Consequently, the controller $K_{H_3}(s)$ does not provide sufficient robustness with respect to noise, motivating the introduction of additional weighting W_4 to limit high-frequency amplification.

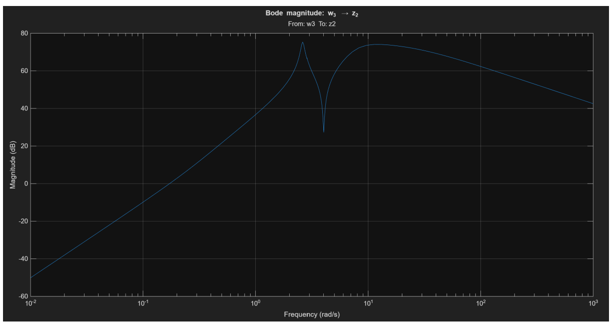

Figure 36: Magnitude of $T_{w_3 \to z_2}$ - Noise perturbation on commanded torque

Since the noise magnitude is of order 10^{-4} while the disturbance magnitude is of order 10^{-3}, the peak gain of the transfer from w_3 to z_2 must remain below 10 to ensure that noise-induced perturbations on the control activity do not dominate the effect of disturbances.

This requirement imposes a bound on the amplification along the noise path through the controller. The weighting function W_4 is therefore tuned to satisfy this constraint while remaining as close as possible to the limit, in order to avoid unnecessary degradation of the other performance objectives. Using $\mathcal{H}_\infty$ synthesis, a value $\gamma_4 = 1.89$ was obtained, indicating that the constraint is globally not satisfied. At least one transfer does not answer to the fixed bound, and we have to check whether it is critical or not.

After analysis, the weighting function $W_4 = 0.78$ effectively limits the noise-induced control activity without introducing critical degradation in the other critical closed-loop transfers, resulting in a well-balanced robustness–performance trade-off (Fig. 37).

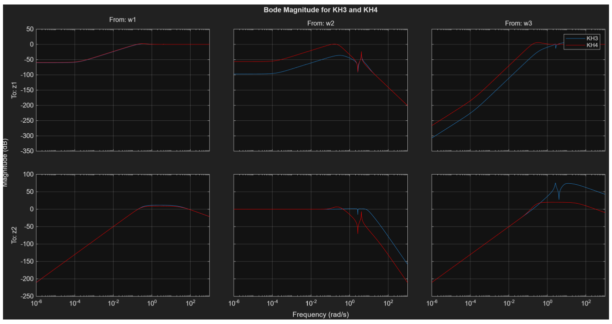

Figure 37: Comparison of Bode Magnitude with K_{H_3} and K_{H_4} controllers

In addition to noise rejection, the robustness of the controller with respect to system uncertainties was also improved, as shown in Fig.38 and Fig.39. However, increasing the severity of the weighting function W_4, for example by setting $W_4 = 15$, leads to a degradation of other performance aspects. In particular, the rejection of torque disturbances is reduced.

This highlights a trade-off between robustness to uncertainties and disturbance rejection, raising the question of whether such a compromise is acceptable in an industrial context.

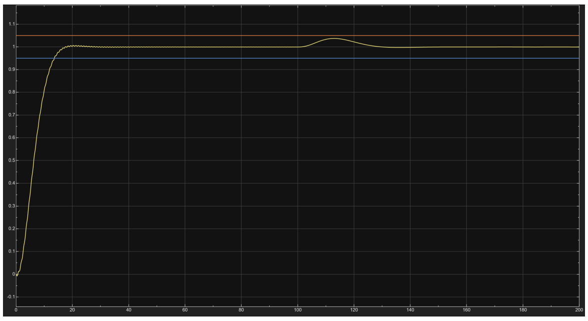

Figure 38: Time Response of the KH3 controller with $r_\alpha = 1.30; r_\omega = 1.30$ for 200s simulation

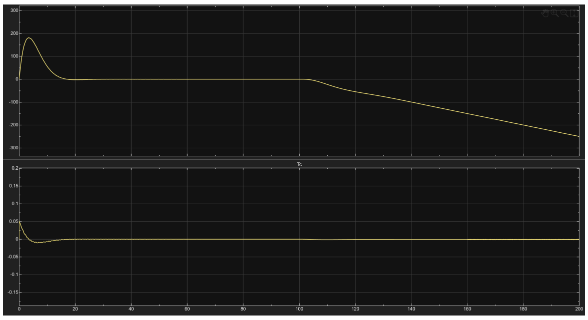

Figure 39: Time evolution of the reaction wheel velocity and the commanded torque for 200s simulation

Since the full H_∞ controller generally has an order higher than the plant, it is not well suited for practical implementation. As a next step, the objective of this lab will be to reduce step by step the order of the computed controller, using **hinfsyn**, while monitoring the closed-loop performance at each reduction stage, in order to find the best trade-off between controller complexity and performance.